

PROG 6212

POE Part 1

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API End Point Plan

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## Detailed Narrative

RaceDay is a web-based event management system designed for South African road running, walking and cycling events. The system will eliminate paper-based registration, spreadsheets and disconnected communication channels, that leave organisers overwhelmed and participants underwhelmed. The system will support two main user roles, **Event Organisers** and **Participants**. Event Organisers will be able to create and manage events, categories and participant results while Participants can browse upcoming events, enter events, track their personal performance history and prepare for their race day.

**Event Organisers** can log in securely and update their own profile information. They can manage events, including **creating, updating, editing** and **deleting** them. **Event Organisers** will capture finishing times for participants, while the system will calculate placement based on the recorded finishing times to reduce manual errors. They will also be able to view all Participants entered in an event. Organiser accounts are created by an authorised organiser/system (Seed data).

Each event must have a **name, description, date, location, distance** and **event type** (run, walk or cycle). Each event can have one or more **categories based on age** (Under 20, Senior) and **distance** (10km, 21km). Events may have an opening and closing date for entries so that the system can determine whether registration is available.

Any user on the website can browse the website for RaceDay events and view general information about different races and race locations.

**Participants** will be able to **self-register** on the web page and receive their **RaceDay unique number**. They can then use their registered login credentials to log in and **update** their own profile. They can browse events and will be able to **enter** suitable event **by selecting a category**. Participants will not be allowed to register for events that they do not qualify for (eg Under 20). Authenticated Participants can **view all of their personal information**.

The system will use a relational SQL Server database to store users, participant and organiser information, events, categories, entries and results. The application will later be accessed through a **RESTful API**, allowing different parts of the system to communicate using **HTTP methods** such as **GET, POST, PUT, PATCH** and **DELETE**. Role-based access control will ensure that **organisers** can **manage events, categories, entries** and **results**, while **participants**

can **manage** their **own profile** and **event entries** but cannot modify organiser-controlled event information.

In later stages of development, the project will include **an ASP.NET Core MVC front end** that communicates with the **API**, image storage using **Azure Blob Storage** and **containerisation** using **Docker**. **GitHub** will be used for **version control, documentation** and **CI/CD** so that changes to the project can be tracked and automatically checked during development. Unit tests will test functionality.

All of this will result in a smooth functioning Web-page for RaceDay events management.

## Business Rules

- A Participant may self register to create a RaceDay account
- A UserName must be unique and each Participant must have a unique RaceDayNumber.
- A Participant may browse events before and after login
- A Participant may enter an event by selecting a category
- Each registration must link one Participant to one Event and one Category
- A Participant should not be able to register in the same event/category more than once
- A participant can view their own registration and track personal results
- A participant may only update their own profile
- A Participant may enter many Events, and an Event may have many Participants
- A Participant may only enter a category for which they meet the eligibility requirements
- A Participant may register for an Event only once. The selected Category must belong to that Event and is recorded as part of the Registration.
- A Participant may only register for an event while registration is open and before the event has been completed
- Each Registration may have zero or one Result.
- An Event must have an event name, description, date, location, distance and event type (run, walk or cycle)
- An Event must have one or more categories (age/distance)
- A Category's minimum age must not be greater than its maximum age.
- An Event's registration closing date must not be before its registration opening date.
- An Organiser must be able to view, create, edit and delete Events
- Any Organiser can edit any event, or category

- An Organiser must manage event categories
- An Organiser must be able to capture race results
- Each Event is linked to one Organiser, although any authorised Organiser may edit an Event or Category.
- A Result may only be recorded for a Participant with a valid registration in that Event
- The system calculates placement according to finishing times within the same Event and Category
- An Organiser must be able to view all event registrations
- A user's authenticated role determines their access permissions
- Only an authorised Organiser may create another Organiser account; initial Organiser accounts may be created through seed data.
- The system will calculate placement according to finishing times per race
- Passwords must never be stored in plain text and must be securely hashed
- Users must be able to end their session and be redirected appropriately

## AI Assistance

ChatGPT (OpenAI) was used as a learning proof reading and support tool during the development of this project.